

Mustafa Ahmed

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EDUCATION

Worcester Polytechnic Institute

Worcester, MA

Bachelor's in Electrical and Computer Engineering

Expected May 2028

- **Relevant Coursework:** Thermodynamics, Atmospheric and Space Environments, Microelectronic Circuits I & II, Introduction to RF Circuit Design, Controls Engineering, Computational Engineering
- **Activities:** Solid State Circuit Society, Muslim Student Association, FIRST Robotics, VEXU Robotics, Chess Club, Demoscene

EXPERIENCE

Manufacturing Engineering Intern

June 2026 – Aug. 2026

Coghlin Companies

Westborough, MA

- Created 6 wire diagrams to support new product designs and manufacturing documentation
- Designed 4 assembly aids and assistive parts to improve production and assembly processes
- Supported 2 new product launch initiatives through assembly, production control, and engineering collaboration
- Applied electrical engineering principles to troubleshoot and improve product functionality

Team Leader

July 2023 – June 2024

FIRST Robotics

Northborough, MA

- Led a 20-member student team to design, prototype, and compete with a 125-lb competition robot
- Designed and fabricated mechanical assemblies including gear trains and linear actuators using Fusion 360, CNC machining, and 3D printing
- Programmed robot control systems and multiple autonomous routines in Java
- Mentored new members in electronics assembly, wiring, programming, and competition strategy

PROJECTS

Vision-Powered Robot Arm | *Arduino, Stepper Motors, Servos, Computer Vision, 3D Printing*

- Designed and built a 4-axis robotic arm using 3 stepper motors, 2 servo motors, and an Arduino microcontroller
- Configured stepper motor drivers and developed Arduino code for coordinated multi-axis motion control
- Integrated computer vision to provide visual input for robotic arm operation
- Modeled and 3D printed 12 structural components and assembled the complete mechanical system

PID-Controlled F1 Model Car | *Arduino, PID Control, Smart Motors*

- Designed and built an Arduino-controlled F1 model car using 2 smart motors and PID feedback control
- Tuned PID parameters to improve vehicle stability and control over a 15-ft test course
- Integrated motor feedback and control logic to achieve consistent vehicle motion
- Successfully transported a stack of 10 blocks without dropping any blocks

FRC Match Strategy Planning App | *Kotlin, Jetpack Compose, REST API*

- Developed a native Android application in Kotlin using Jetpack Compose for FRC match scouting and alliance strategy
- Built 4 major application screens with support for autonomous and teleoperated match planning
- Created an interactive field map supporting 3 draggable robot markers, path visualization, scoring zones, and live notes
- Integrated The Blue Alliance REST API to retrieve team statistics, event schedules, and historical match data

Educational FRC Subsystem Prototypes | *Fusion 360, Arduino, 3D Printing*

- Designed and built 3 tabletop FRC subsystem prototypes including drivetrains, elevators, and shooter mechanisms
- Modeled scaled components in Fusion 360 and fabricated parts using 3D printing
- Integrated DC motors and Arduino-based controllers to demonstrate gear ratios and closed-loop control

TECHNICAL SKILLS

Programming: Java, C/C++, Python, Kotlin, HTML

Hardware & Controls: Arduino, Embedded Systems, Computer Vision, PID Control, Stepper Motors, Servo Motors, Motor Drivers, Electrical Wiring, Soldering, Autonomous Programming, FRC Robot Design

Design & Manufacturing: Fusion 360, CAD, 3D Printing, CNC Machining